

Shamim Rahim Refat

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Shamim Rahim Refat | Shamim Rahim Refat

SUMMARY

Aspiring enthusiast in Machine Learning and Artificial Intelligence, focusing in computer vision and natural language processing. Enthusiastic about contributing to dynamic teams and continuously advancing my skills and knowledge through hands-on experience and learning.

WORK EXPERIENCE

Technohaven Company Ltd.

June 2025 - Present

Jr. Software Engineer (AI/ML)

- Working on multi-Agent, bilingual (Bangla & English) AI chatbots using Python, PyTorch, Open AI API, LangGraph, LangChain, FastAPI and Streamlit for real-time fintech application and banking assistance.
- Developed an AI-driven fintech (digital banking) system integrating transaction analysis and customer support.
- Created RAG pipelines for document-based question answering to improve financial insights.
- Worked on advanced computer vision pipelines for real-world applications, focusing on image-based understanding and visual reasoning tasks.
- Developed and optimized virtual try-on systems using deep learning techniques, emphasizing pose alignment, garment warping, and visual realism.
- Conducted large-scale data retrieval and analysis workflows using transfer learning, Vision Transformers (ViTs), and convolutional neural networks.
- Designed and trained object detection models using YOLO architectures for structured visual data extraction.
- Integrated Optical Character Recognition (OCR) systems to extract and process textual information from complex image-based datasets.
- Collaborated with cross-functional teams to deliver responsible, AI-powered banking solutions.

A.R.i.S.E Research Lab

January 2025 - Present

Research Assistant

- Contributed to cutting-edge research in Deep Learning, Computer Vision and Explainable AI, resulting in the publication of three peer-reviewed conference papers and two journal articles (Q1 and Q2).
- Actively involved in collaborative research, model development, and dataset curation.
- Mentored and supervised undergraduate students and junior lab members in project design, model implementation, and academic writing.

EDUCATION

2020 - 2025	B.Sc. in CSE at Ahsanullah University of Science and Technology	(GPA: 3.096/4.0)
2017 - 2019	Higher Secondary School Certificate at Notre Dame College	(GPA: 5.0/5.0)
2007 - 2017	Secondary School Certificate at Ideal School and College	(GPA: 5.0/5.0)

PUBLICATIONS

- Bidyarthi Paul, Faika Fairuj Preotee, Shuvashis Sarker, **Shamim Rahim Refat**, Shifat Islam, Tashreef Muhammad, Mohammad Ashraful Hoque and Shahriar Manzoor, "ANCHOLIK-NER: A Benchmark Dataset for Bangla Regional Named Entity Recognition", Accepted in PLOS One (Q1 Journal), DOI: [10.1371/journal.pone.0342786](https://doi.org/10.1371/journal.pone.0342786).
- Shamim Rahim Refat**, Ziyen Shirin Raha, Shuvashis Sarker, Faika Fairuj Preotee, MD. Musfikur Rahman, Tashreef Muhammad and Mohammad Shafiqul Alam, "VR-FuseNet: A Fusion of Heterogeneous Fundus Data and Explainable Deep Network for Diabetic Retinopathy Classification", Accepted in Biomedical Materials and Devices (Q2 Journal), Preprint: [arXiv](https://arxiv.org/abs/2501.00000).

- Shuvashis Sarker, **Shamim Rahim Refat**, Faika Fairuj Preotee, Tanvir Rouf Shawon and Raihan Tanvir, "Comprehensive Lung Disease Detection Using Deep Learning Models and Hybrid Chest X-ray Data with Explainable AI", 2024 27th International Conference on Computer and Information Technology (ICCIT), DOI: [10.1109/ICCIT64611.2024.11022605](https://doi.org/10.1109/ICCIT64611.2024.11022605).
- Shuvashis Sarker, **Shamim Rahim Refat**, Faika Fairuj Preotee, Shifat Islam, Tashreef Muhammad and Mohammad Ashraful Hoque, "An Exploratory Approach Towards Investigating and Explaining Vision Transformer and Transfer Learning for Brain Disease Detection", 2024 27th International Conference on Computer and Information Technology (ICCIT), DOI: [10.1109/ICCIT64611.2024.11022039](https://doi.org/10.1109/ICCIT64611.2024.11022039).
- Faika Fairuj Preotee, Shuvashis Sarker, **Shamim Rahim Refat**, Tashreef Muhammad and Shifat Islam, "An Approach Towards Identifying Bangladeshi Leaf Diseases through Transfer Learning and XAI", 2024 27th International Conference on Computer and Information Technology (ICCIT), DOI: [10.1109/ICCIT64611.2024.11022335](https://doi.org/10.1109/ICCIT64611.2024.11022335).
- Shuvashis Sarker, **Shamim Rahim Refat**, Faika Fairuj Preotee, Tashreef Muhammad, "Advanced CNN and Explainable AI Based Architecture for Interpretable Brain MRI Analysis", Proceedings of the 3rd International Conference on Computing Advancements. ICCA '24, DOI: [10.1145/3723178.3723220](https://doi.org/10.1145/3723178.3723220).

SKILLS

Programming Languages	Python, C++, Java, Dart
Deep Learning Frameworks	Tensorflow, Keras, PyTorch, scikit-learn
LLM Application Frameworks	LangChain, LangGraph, LangSmith
Databases	ChromaDB, Microsoft SQL Server, MySQL
Others	FastAPI, Streamlit, Gradio, Git

REFERENCES

- **Dr. Mohammad Shafiul Alam**
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